A Survey of Robotic Arm Systems Approaching Embodied Intelligence

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Robotic arms serve as essential physical interfaces for robotic interaction with the environment in the physical world. Recent advances in robotic sensing and learning-based methods have substantially expanded the capabilities of robotic arms, bringing such systems closer to embodied intelligence. This article therefore examines robotic arm systems approaching embodied intelligence by focusing on the co-evolution of embodiment infrastructure, data ecosystems, learning paradigms, and evaluation protocols. It first reviews physical embodiment and software infrastructure that make robotic arm manipulation feasible and reproducible. It then investigates how embodied manipulation data are organized, collected, refined, and reused. It further reviews how manipulation capability is acquired and organized through the coordination of cognitive planning and policy learning, and how it is deployed in real-world settings. It also examines evaluation practices in relation to task demands, data conditions, generalization requirements, and deployment contexts. Finally, it identifies open challenges for more systematic progress in robotic arm systems toward embodied intelligence.

CCS Concepts: • **Computer systems organization → Robotic autonomy**; *Robotic control*; External interfaces for robotics.

Additional Key Words and Phrases: Robotic arm systems, data, learning, embodiment, evaluation

ACM Reference Format:
Wei Jiang, Sen Wang, Jinyu Zhan, and Xingzhi Zhou. 2026. A Survey of Robotic Arm Systems Approaching Embodied Intelligence. *ACM Comput. Surv.* XXX, XXX, Article XXX (May 2026), 35 pages. https://doi.org/XXXXXXX.XXXXXXX

1 Introduction

Robotic arm systems are widely used in industrial automation, surgical assistance, and household service [110, 113, 141]. Early research focused on platform construction, kinematic and dynamic modeling, motion planning, control, and simulation, establishing the foundations for reliable robotic arm manipulation. Building on these foundations, advances in robotic sensing and learning-based methods have supported the development of robotic arm systems toward embodied intelligence. As summarized in Figure 1, this evolution can be traced from foundational middleware and simulation platforms, such as ROS/ROS2 and MuJoCo [116, 161], to scalable data-collection and multi-robot demonstration systems, such as RoboTurk, RoboNet, and Open X-Embodiment (OXE) [37, 119, 130].

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ACM 1557-7341/2026/5-ARTXXX
https://doi.org/XXXXXXX.XXXXXXX

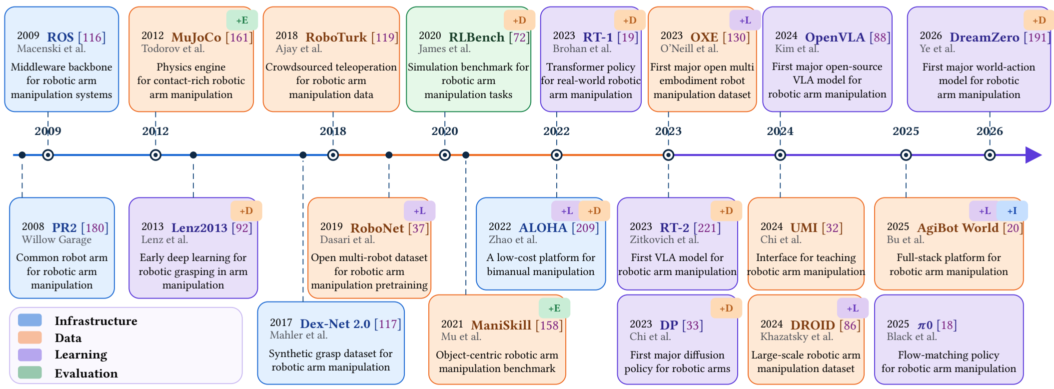

Fig. 1. **Chronological Overview of Robotic Arm Systems' Evolution.** Milestones are colored by their primary layer in the manipulation stack.Upper-right tags indicate secondary layer associations, with +I, +D, +L, and +E denoting Infrastructure, Data, Learning, and Evaluation.

It then extends to modern imitation-learning-based policy methods such as RT-1 and Diffusion Policy [19, 33], and more recently to vision-language-action (VLA) models such as RT-2, OpenVLA, and π_0 [18, 88, 221]. Existing works have covered unified abstractions for robotic manipulation [11], vision-language-action models for real-world robotics [84], object-centric embodied learning [213], and broader embodied manipulation systems based on LLMs and world models [108]. While these works provide valuable perspectives on individual dimensions, they rarely examine how these dimensions co-evolve and interact across the full system stack. As a supplementary, this article organizes this cross-layer evolution across four interrelated dimensions. First, it reviews embodiment infrastructure, covering the physical embodiment and software foundations that make robotic arm manipulation feasible and reproducible. Second, it examines the data ecosystem, including how embodied manipulation experience is collected, organized, refined, reused, and incorporated into downstream learning and deployment pipelines. Third, it reviews learning paradigms, focusing on how manipulation capabilities are acquired and structured through cognitive planning, policy learning, integration models, and deployment pipelines. Fourth, it discusses evaluation protocols, relating system performance to task execution, data conditions, generalization requirements, and deployment contexts.

Contributions. First, this paper provides a system-level perspective for examining robotic arm systems toward embodied intelligence. It treats robotic arm manipulation as an integrated and co-evolving system rather than a collection of isolated advances in hardware, data, models, and benchmarks. This perspective clarifies how progress in the field depends on the coordination of infrastructure, data production, learning, and evaluation. Second, it provides a cross-layer organization of literature on robotic arm systems. It situates existing works within the manipulation stack and uses structured taxonomies to clarify their relations and interactions. This organization connects embodiment foundations, data acquisition and refinement, learning and deployment, and evaluation into a coherent account of the manipulation stack. Third, it examines recurring bottlenecks in robotic arm systems from a system-level perspective. It traces how limitations originating in earlier layers propagate and accumulate across data, learning, and evaluation. This analysis clarifies why challenges in robotic arm systems cannot be fully addressed within isolated components. Through this systematic account, we aim to provide a clearer basis for understanding the progress and limitations of robotic arm systems.

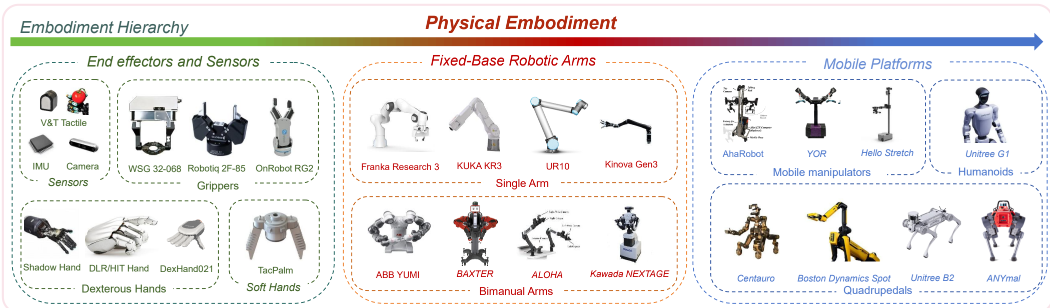

Fig. 2. Overview of physical embodiments for robotic arm manipulation. The figure covers end-effectors, sensors, stationary platforms, and mobile platforms.

2 Embodiment Infrastructure for Robotic Arm Systems

For robotic arm platforms, the foundations of manipulation lie in two aspects of embodiment. Physical embodiment defines how interaction is physically structured for robotic arm manipulation through contact, sensing, action, and platform constraints. Software Foundation makes these physically grounded capabilities operational by supporting data capture, communication, low-level control, and the formalization of robotic experience for downstream learning.

2.1 Physical Embodiment

Physical embodiment is the source of embodiment constraints in robotic arm manipulation. It determines how contact can be made, sensed, and transformed into controllable behavior. We begin with end-effectors and sensors, then turn to robotic arms as the platforms that organize multi-DoF motion into executable manipulation, and finally discuss how arms are integrated into larger mobile and whole-body systems. Figure 2 provides an overview of the physical embodiment hierarchy considered in this section. As illustrated in the figure, physical embodiment spans end-effectors and sensors, fixed-base robotic arms, and mobile platforms.

2.1.1 End-effectors and Sensors. End-effectors provide the physical interface through which robotic arms interact with objects. Their design imposes local contact constraints, determining how contact can be established, stabilized, adjusted, and released during manipulation. Jaw grippers, such as Robotiq 2F-85, favor stable and mechanically constrained contact, making grasping, alignment, and release easier to control and predict. Dexterous hands, such as DLR Hand [21], support rich in-hand manipulation by allowing contact points and forces to change during execution, but this flexibility also makes their behavior harder to model as a small set of repeatable contact patterns. Soft hands, such as SpiRobs [174], use compliance to absorb part of the variations in contact conditions, object shape, and grasp alignment within the hardware itself. Sensors impose perceptual constraints on manipulation by determining what task-relevant states can be observed during execution. The central task of sensors is to ensure the information required at different stages of interaction is observable. Before contact, the dominant uncertainties concern object geometry, pose, and scene layout, so visual sensing is often sufficient for coarse localization, alignment, and approach. During contact, however, task-relevant information becomes increasingly local and interaction-dependent, including slip, deformation, insertion error, and force misalignment. These states are often critical to task success, yet they are only weakly reflected in external visual observations. For this reason, tactile, force, and proprioceptive sensing become critical once task-relevant information is expressed primarily through contact itself. Sensors such as GelSight [98], DIGIT [89], and Inertial Measurement

Units complement external vision by capturing local contact geometry, contact transients, and motion signals that are difficult to observe from camera views alone. Contact space and observation space define the physical and informational boundaries of robotic arm manipulation. The resulting contact events and sensory observations form essential components of the data support discussed in Section 3.

2.1.2 Fixed-base Robotic Arms. In common fixed-base arm manipulation settings, controllable multi-DoF motion is used to position end-effectors and sensors during task execution. Within the contact and observation spaces defined above, motor commands are typically structured into manipulation sequences, including approach, alignment, contact, transport, and release. Through active motion and contact, these sequences induce changes in object pose, contact relations, task state, and sensory observations. In single-arm settings, such as KUKA LBR iiwa [123], the action structure is relatively localized. It is usually organized around a compact set of local objectives, including stable grasping, object alignment, and local contact transition. Because the execution structure is centered on one arm and one primary contact interface, motion sequences and state transitions in single-arm settings are relatively easier to collect, model, and evaluate. Consequently, single-arm systems are the dominant setting for teleoperation, policy benchmarking, and high-precision manipulation. In bimanual settings, such as PR2 [180], manipulation requires coordination across multiple arms. When multiple manipulators act on the same object, the task is better characterized as coordinated object transformation under shared kinematic and force constraints. This coordination enables tasks that require force regulation across arms or the handling of large, heavy, articulated, or deformable objects. As a result, bimanual systems are increasingly used in assembly, tool use, and human-robot collaboration. These manipulation settings arise from practical task demands, but they also determine how motion sequences, contact relations, and state transitions are organized for data collection, policy learning, and evaluation. Motion sequences and the state transitions they induce form an important part of the data support will be discussed in Section 3.

2.1.3 Mobile Platforms. Mounting robotic arms on mobile platforms expands manipulation beyond fixed-base settings. In wheeled mobile manipulators such as Mobile ALOHA [45] and Stretch [85], arm motion must be coordinated with base motion, navigation constraints, and changing viewpoints. In legged platforms and humanoids, manipulation is further coupled with balance, posture regulation, and whole-body support conditions. Overall, mobile and whole-body platforms place robotic arms in a broader execution context, requiring manipulation to be planned and executed together with mobility, sensing, and body stability.

2.2 Software Foundation

The Software Foundation of robotic arm manipulation turns physical embodiment into executable and recordable processes. During runtime, it captures sensor and actuator signals, transports data across distributed components, and sustains low-level control. It also supports the logging, synchronization, and structuring of interaction data for downstream learning. In this way, these mechanisms turns embodiment constraints into executable, recordable, and reusable system processes.

2.2.1 Modern Robotics Middleware and Operating Systems. The software stack in robotic arm manipulation systems provides the infrastructure for capturing, transmitting, and organizing embodied signals and data. It ensures that sensor and control signals are captured with sufficient temporal fidelity, transmitted without major bottlenecks, and managed at scale across heterogeneous hardware platforms and computational resources. We organize this stack into three tiers, as summarized in Table 1. The Capture Layer (L0) ensures the temporal determinism required for reliable signal acquisition from sensors and actuators. Industrial stacks rely on real-time operating system (RTOS)

Table 1. **Three-tier software architecture of robotic arm manipulation for embodied intelligence.** The software stack is organized into three layers. L0 is the Capture Layer for hard real-time motor control with microsecond-level determinism. L1 is the Transfer Layer for high-throughput, low-latency inter-process and inter-device communication. L2 is the Orchestration Layer for distributed orchestration across robots, edge devices, cloud resources, and fleets. The Profile column reports interrupt or IPC latency for L0 and L1 systems, and deployment or orchestration characteristics for L2 systems.

Framework	Type	Core Mechanism	Key Contribution	Profile	Scope
			L0: Capture Layer		
VxWorks [188]	Monolithic	Priority Preemptive	Deterministic RTOS kernel with priority-based preemptive scheduling, designed for low-latency and low-jitter execution in control systems.	<10 μs	Industrial
QNX [60]	Microkernel	Spatial Isolation	Microkernel RTOS that isolates drivers and system services, improving fault containment while preserving deterministic scheduling.	<5 μs	Safety-critical
			L1: Transfer Layer		
ROS 2 [116]	Decentralized	DDS/RTPS	Middleware built on DDS/RTPS, providing configurable QoS policies for reliability, deadline, and liveliness in distributed robotic systems.	1–10 ms	General Robotics
Dora-rs [206]	Dataflow	Apache Arrow	Rust-based dataflow runtime using Apache Arrow and shared memory for low-latency, zero-copy exchange across modular robotic processes.	<50 μs	Low-latency AI
			L2: Orchestration Layer		
Zenoh-Flow [13]	DAG	Zenoh Protocol	Communication and query protocol that unifies pub/sub, storage, and distributed computation across devices ranging from embedded nodes to edge and cloud systems.	Cloud	Distributed
FogROS2 [71]	Cloud	ROS 2 Graph	ROS 2 extension which enables selected components of a robotic graph to be offloaded while retaining local reactive control.	Hybrid	Cloud robotics

kernels such as VxWorks [188] and QNX [60] for bounded interrupt latency, predictable scheduling, and fault isolation in low-level sensing and actuator control. The Transfer Layer (L1) moves captured data between processes, devices, and model components. ROS 2 provides the baseline connectivity through decentralized Data Distribution Service (DDS) [116], while the growing cost of moving multimodal embodied signals and data has driven zero-copy designs such as Dora-rs [206]. The Orchestration Layer (L2) organizes data flows across robots, edge devices, and cloud services at the system level. Zenoh-Flow and FogROS2 are designed with deployment, routing, and compute offloading as core system concerns [13, 71], while RoboOS couples multimodal planning to reusable skill libraries across heterogeneous embodiments [157].

2.2.2 Control Execution and Data Management. During execution, robotic arms rely on low-level planning and control to execute motion and contact behavior. For end-effector interaction, impedance control enables compliant behavior. This compliance helps maintain stable contact, tolerate small pose errors, and reduce excessive interaction forces. These properties make impedance control useful for assembly, insertion, tool use, and other contact-rich manipulation tasks. For fixed-base arms, planning and control convert operational objectives into joint motions and torques. This process relies on kinematic mapping, rigid-body dynamic compensation, and joint and collision constraints. For mobile platforms and humanoids, manipulation is further coupled with locomotion, balance, and whole-body motion. MPC (Model Predictive Control) and WBC (Whole-Body Control) provide complementary control formulations for coordinating motion, contact, balance, and whole-body constraints in mobile and humanoid manipulation [83]. Beyond execution, software

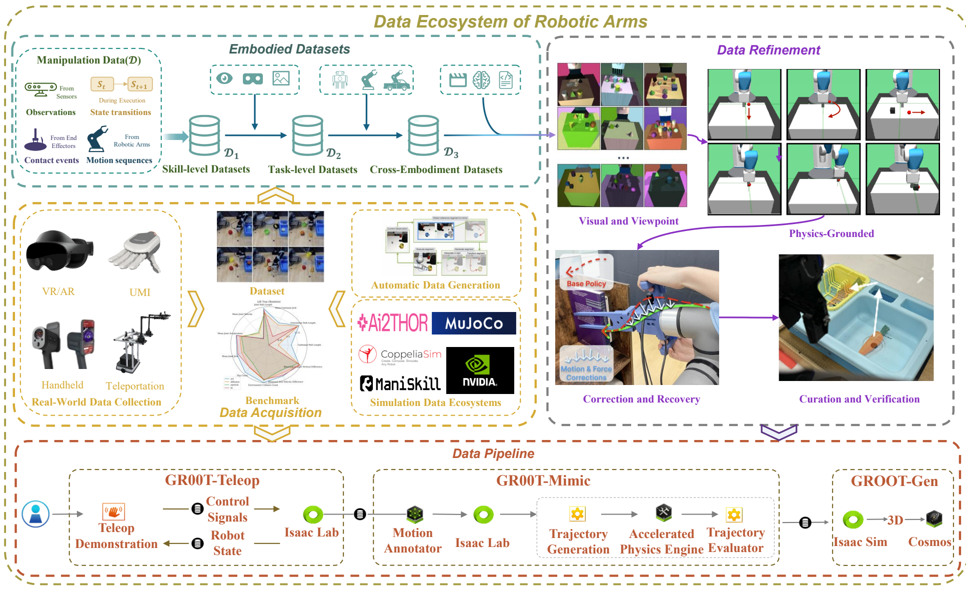

Fig. 3. Overview of the embodied data ecosystem, organized into embodied datasets, data acquisition, data refinement, and data pipelines. The graph shows how data are collected, structured, refined, and incorporated into downstream learning and deployment pipelines.

infrastructure also supports how interaction traces are recorded, synchronized, and structured for later reuse. In practice, datasets use different storage and schema conventions to preserve embodied interaction records for downstream learning. Some containers such as Hierarchical Data Format (HDF5) store observations, actions, rewards, timestamps, and metadata, while other schemas such as Reinforcement Learning Datasets (RLDS) organize these records into trajectories.

3 Data Ecosystem for Robotic Arm Systems

Having clarified how embodiment constrains robotic arm manipulation through contact, observation, and execution, we now turn to the data ecosystem through which these constraints are captured, formalized, and made learnable. In this context, data refer to the recorded traces of embodied manipulation rather than generic observations or files. We denote such data by $\mathcal{D}$, which records these traces as the robot engages with the world. Concretely, $\mathcal{D}$ comprises four components: **motion sequences** generated by the robotic arm, **sensory observations** provided by onboard or external sensors, **contact events** arising through the end-effector, and the resulting **state transitions** induced during execution. From the perspective of embodied intelligence, progress in the data ecosystem can be understood in terms of how $\mathcal{D}$ is organized, collected, refined, and reused across tasks, scenes, and embodiments. Organized around $\mathcal{D}$, this chapter first reviews embodied datasets in Section 3.1, and then examines the broader processes through which exec experience is produced and operationalized, including data acquisition in Section 3.2, data refinement in Section 3.3, and large-scale data pipelines in Section 3.4. These components of $\mathcal{D}$ are also implicitly encode reusable structure about perception, action, contact, and state change, which later serves as knowledge support $\mathcal{K}$ for Section 4. Together, these components will be used to provide a structured account of how data support is formed under embodiment constraints for robotic arm manipulation, as summarized in Figure 3.

Table 2. **Chronological Overview of Skill-level Datasets.** Evolution reflects the shift from planar grasping rectangles to dense 6-DoF poses, dexterous hands, language-grounded reasoning, and diverse motor primitives (pushing, articulation, deformable manipulation, assembly).

Dataset	Year	Skill Type	Scene Complexity	Objects	Domain	Scale	Modality	Lang.
			Early Geometric Grasping					
Extended Cornell [92]	2013	Rect. Grasp (2D)	Single Object	240	Real	1k imgs	RGB-D	×
Dex-Net 2.0 [117]	2017	Parallel-Jaw Grasp	Single Object	1.5k	Sim	6.7M grasps	Depth	×
			Dense 6-DoF Grasping					
GraspNet-1B [41]	2020	6-DoF Grasp	Heavy Clutter	88	Real	97k imgs (1.2B grasps)	RGB-D	×
GraspClutter6D [10]	2025	6-DoF Grasp	Cluttered	200	Real	52k imgs (9.3B grasps)	RGB-D	×
			Dexterous Grasping					
DexGraspNet 2.0 [200]	2024	Dexterous Grasp	Multi-Object	5,355	Sim	427M grasps	Point Cloud	×
Dex1B [190]	2025	Dexterous Grasp	Single Object	1M	Sim	1B grasps	Point Cloud	×
			Language-Grounded Grasping					
Grasp-Anything [163]	2023	Rect. Grasp + Push	Multi-Object	3M	Syn	1M imgs (600M grasps)	RGB	✓
ReasoningGrasp [80]	2024	6-DoF Grasp	Multi-Object	64	Real	99M grasps	RGB-D	✓
			Diverse Motor Primitives					
Omnipush [15]	2019	Pushing (Planar)	Single Object	250	Real	62.5k pushes	RGB-D	×
PartNet-Mobility [182]	2020	Articulated (Open/Close)	Single Object	2,346 models	Sim	2,346 articulated assets	3D / PC	×
FurnitureBench [57]	2023	Assembly (Precise)	Tabletop	Furniture	Sim+Real	5.1k demos (220 hrs)	RGB	×

3.1 Embodied Datasets

Existing robotic arm manipulation datasets show a clear progression toward broader forms of support for learning: from capturing local interaction structure, to recording temporally extended task execution, and ultimately to supporting transfer across heterogeneous embodiments. We organize this progression into three tiers of embodied datasets: skill-level datasets ($\mathcal{D}_1$), which capture local interaction structure; task-level datasets ($\mathcal{D}_2$), which record how manipulation unfolds over extended execution; and Cross-Embodiment datasets ($\mathcal{D}_3$), which support transfer across embodiments at scale. This tiered view reflects the increasingly demanding data support required for embodied intelligence. The following subsections review skill-level datasets in Section 3.1.1, task-level datasets in Section 3.1.2, and Cross-Embodiment datasets in Section 3.1.3.

3.1.1 $\mathcal{D}_1$: Skill-level Datasets. The development of Skill-level Datasets ($\mathcal{D}_1$) reflects a shift from grasp-centric annotations toward richer forms of local interaction data, including 6-DoF grasp synthesis, dexterous hand configurations, pushing, articulation, and assembly. Table 2 lists representative datasets along this tier. Early grasping benchmarks such as Cornell [92] labeled graspable regions as 2D oriented rectangles, covering vertical parallel-jaw approaches. Dex-Net 2.0 [117] generated 6.7 million synthetic grasp examples labeled with analytic grasp metrics, enabling large-scale grasp learning without collecting training data through physical trials. GraspNet-1B [41] annotated 1.2 billion 6-DoF grasps across cluttered real-world scenes, and GraspClutter6D [10] extended this to denser clutter configurations. For multi-fingered hands (>20 DoF), DexGraspNet 2.0 [200] synthesized 427M grasps via diffusion-based generation (90.7% real-world success), and Dex1B [190] reached one billion grasps. Tactile modalities complement vision: T-DEX [51] showed that 2.5 hours of self-supervised tactile play data improves dexterous task success by 1.7× over baselines with only vision, capturing slip and reorientation signals invisible to cameras. Beyond grasping, Omnipush [15] records planar pushing dynamics across 250 objects, PartNet-Mobility [182] provides part interaction data for articulated objects, and FurnitureBench [57] supplies 5,000 teleoperated assembly trajectories with a paired simulator.

3.1.2 $\mathcal{D}_2$: Task-level Datasets. Task-level datasets ($\mathcal{D}_2$) evolve by extending isolated motor primitives into trajectories aligned with language instructions, gradually increasing semantic coverage,

Table 3. **Representative task-level embodied datasets collected from real-world physical trajectories.** Traj. denotes the number of recorded demonstrations or episodes; V denotes the number of camera views available; Tasks denotes the number of distinct task types; Scenes denotes the number of distinct environments or scene settings; Lang. indicates whether language annotations or task instructions are provided; Depth indicates whether depth observations are provided; Calib. indicates whether camera calibration or equivalent calibration metadata is available; Form. indicates the primary release format; Platform denotes the robot platform used for data collection; Collection Mode summarizes the acquisition protocol. In the table, Teleop. = teleoperation, and H/S Hybrid = a combination of human demonstrations and scripted data collection.

Dataset	Year	Traj.	V	Tasks	Scenes	Lang.	Depth	Calib.	Platform	Collection	Form.
MIME [147]	2018	8.3k	3	20	1	×	✓	×	Baxter	Teleop.	Raw
RoboTurk [119]	2018	2.1k	2	3	1	×	✓	×	Panda	Teleop.	HDF5
MT-Opt [81]	2021	800k	1	12	1	×	×	×	Google Arm	Mixed	TFDS
BC-Z [73]	2021	26k	1	100	1	✓	×	×	EDR	Teleop.	TFDS
RT-1 Dataset [19]	2022	130k	1	700+	2	✓	×	×	EDR	Teleop.	TFDS
Dobb-E [145]	2023	1.5k	1	109	22	×	✓	×	Stretch	iPhone	TFDS
RoboSet [17]	2023	28.5k	4	38	Var.	✓	×	×	Panda	Mixed	HDF5
BridgeData V2 [164]	2024	60k	4	13	24	✓	✓	×	WidowX	H/S Hybrid	Raw
DROID [86]	2024	76k	3	84	564	✓	✓	✓	Panda	Teleop.	RLDS
FMB [114]	2025	22k	4	2	1	✓	✓	✓	Panda	Mixed	Raw

environmental diversity, and calibration quality across collection sites. Table 3 lists representative datasets along this tier. MT-Opt [81] collected 800k trajectories through autonomous exploration but used task IDs to index actions, without language supervision. BC-Z [73] introduced language-conditioned supervision by pairing 26k trajectories with 100 semantic verbs, RoboSet [17] further provides single-arm demonstrations across multiple manipulation skills and task instances, and RT-1 Dataset [19] extended this to 130k trajectories spanning 700+ instructions. However, despite their growing scale, these early datasets were collected in a limited range of settings within one or a few research institutions, which restricted environmental diversity. BridgeData V2 [164] widened scene coverage to 24 environments via collection in several locations, DROID [86] coordinated 18 institutions across 564 scenes with calibrated extrinsics, and Dobb-E [145] accessed 22 domestic environments using only an iPhone mounted on a Stretch robot. FMB [114] added force-torque calibration across 22k contact-rich assembly trajectories.

3.1.3 $\mathcal{D}_3$: Cross-Embodiment Datasets. Cross-Embodiment datasets ($\mathcal{D}_3$) organize robot experience across platforms into shared data support for cross-embodiment learning, with increasing emphasis on alignment, standardization, and multimodal consistency. Table 4 summarizes representative datasets in this tier. RoboNet [37] established an early multi-robot dataset with 15 million video frames collected across seven robot platforms. OXE [130] further scaled this tier with over one million real-robot trajectories spanning 22 robot embodiments. As this tier scales across platforms, tasks, and modalities, standardization and multimodal alignment become necessary for turning heterogeneous trajectories into reusable data support. ARIO [175] provides a unified large-scale dataset with approximately three million episodes and consistent timestamp alignment across heterogeneous sources, while RH-20T [42] contributes over 110k contact-rich manipulation sequences with synchronized visual, force, audio, and action signals. RoboMIND [178] extends this tier toward standardized multi-robot task datasets, pairing 107k trajectories across four robot types with failure annotations, and RoboMIND 2.0 [63] scales this to over 310k trajectories across six robot platforms and 739 tasks. AgiBot World Beta [20] further pushes this tier toward large-scale dual-arm and dexterous manipulation, providing over one million trajectories from 100 robots across 100+ real-world scenarios. Some pipelines also incorporate human-centric video sources such as Ego-Exo4D [49] to supplement robot trajectories with richer visual and interaction priors.

Table 4. **Cross-Embodiment Datasets for Multi-Robot Manipulation.** # Traj. = trajectory count; # Embod. = distinct robot platforms/configurations when explicitly reported; Embod. Types: S = single-arm, B = bimanual, M = mobile manipulator, H = humanoid, D = depth, F = force/torque, A = audio; Lang. = language supervision; Action: EE = end-effector, Emb. = embodiment-specific, WB = whole-body; Alignment: how multi-robot differences are reconciled.

Dataset	Year	# Traj.	# Embo.	Types	Source	Modalities	Lang.	Action	Alignment
RoboNet [37]	2019	162k	7	S	Scripted	RGB	×	EE	Interface-level
RH-20T [42]	2023	110k	7	S	Teleop	RGB, D, F, A	✓	Emb.	Calibration-level
OXE [130]	2023	1M+	22	S, B, M	Aggregated	RGB	✓	EE	Schema-level
ARIO [175]	2024	3M	N/A	S, B, M	Aggregated	RGB, D	✓	Emb.	Temporal
AgiBot World [20]	2025	1M+	100+	B, H	Teleop	RGB, D, F	✓	WB	Pipeline-level

3.2 Data Acquisition

The previous section distinguished dataset tiers by the support they provide. However, acquiring such support is not straightforward. Unlike generic datasets, embodied manipulation data need to record manipulation sequences under the constraints of observation space and contact space. Collecting data that is both physically grounded and rich enough to support learning at scale is difficult. This section therefore examines three main routes by which embodied interaction is converted into $\mathcal{D}$ through real-world collection in Section 3.2.1, simulation ecosystems in Section 3.2.2, and automatic data generation in Section 3.2.3.

3.2.1 Real-world Data Collection. Since real-world collection is conducted under concrete hardware and scene constraints, its efficiency and coverage are constrained by the robotic arm platform, end-effectors, and execution environments. Recent researched can be regarded as a gradual effort to weaken this coupling while retaining useful observation, contact, and execution structure. Direct teleoperation on the deployment robot provides a straightforward way to acquire such data, and it forms the basis of many early approaches. RoboTurk [119] built a teleoperation interface that enabled demonstrations to be collected from geographically distributed users over the internet. ALOHA [45, 209] introduced a low-cost teleoperation setup for bimanual and later mobile whole-body manipulation, making long-horizon and contact-rich data collection more accessible. Teleoperation preserves the coupling between observation space, contact space, and arm-centric execution most faithfully, but it is also tightly constrained by hardware access and human labor. For this reason, later work moves toward more portable and reusable data-collection interfaces, so that demonstrations can be transferred more easily across robot embodiments and deployment setups. UMI [32] used handheld grippers and consumer cameras to capture demonstrations in everyday environments and then retargeted them offline to the target robot platform, while DexCap [166] pushed this approach toward dexterous manipulation by recording high-dimensional human hand motion for transfer to multi-fingered robotic manipulators. Even so, these interfaces are limited to narrower robotic arms and still rely on teleoperation. This motivates a further step toward egocentric and video-based supervision by retargeting human demonstrations back into robotic arm execution. EgoMimic [82] used egocentric human video to enable large-scale data capture without requiring robot hardware during collection, and AV-ALOHA [34] showed that such data become more useful when the demonstrator's viewpoint is perceptually aligned with the robot's onboard observation space.

3.2.2 Simulation Data Ecosystems. Compared with real-world collection, simulation weakens the dependence of data acquisition on specific robotic arm platforms, end-effector and sensor setups, and scene conditions. This makes it easier to define different structured representations of $\mathcal{D}$ for robotic arm manipulation under more explicit assumptions. In this section, MuJoCo, SAPIEN, and NVIDIA

Table 5. **Simulation infrastructures for embodied manipulation.** Platforms are grouped into three main ecosystems (MuJoCo, SAPIEN, Isaac) and specialized platforms for specific task categories or physical regimes.

Simulator	Features	Capabilities & Modalities	Key Benchmarks & Datasets
		I. The Core Simulation Ecosystems	
MuJoCo	Stable analytic contact solver; MJX hardware acceleration; Differentiable dynamics.	**Physical Aligned Data**: Trajectories for fine-grained motor control. **Modalities: Proprioception, Contact forces, Kinematics, RGB.**	**Yu et al.** [193], **LIBERO** [106], [126]
Isaac Suite	PhysX5;RTX rendering; Generative synthesis.	**Generative Scalable Data**: Large-scale synthetic and high-DoF control. **Modalities: RGB-D, LiDAR, Segmentation, Bounding boxes, Force/Torque, Event camera.**	**Isaac Lab** [122], **Behavior-1k** [93], **Arnold** [47]
SAPIEN	High-fidelity articulated dynamics; Part mobility; Vulkan rendering.	**Semantic Interaction Data**: Interactions across articulated categories and reasoning over articulated parts. **Modalities: RGB-D, Masks, Contact forces, Articulated object states.**	**ManiSkill3** [158], **SIMPLER** [100], **RoboTwin** [125]
		II. Specialized Task Platforms	
PyBullet / CoppeliaSim	Lightweight legacy physics; Stable IK solvers; Hardware and ROS abstraction.	**Algorithmic Baseline Data**: Logic planning and action primitives. **Modalities: RGB-D, Joint states, Contact forces, Proximity sensors, Force/Torque.**	**CALVIN** [121], **RLBench** [72]
Habitat / AI2-THOR	Procedural indoor scene generation; Scalable visual navigation environments.	**Embodied Navigation Data**: Semantic search and indoor reasoning. **Modalities: RGB-D, Segmentations, Object states, Agent pose, Audio.**	OpenEQA [118], ALFRED [151]

Isaac are treated as three representative directions because they align naturally with different tiers of $\mathcal{D}$ for robotic arm manipulation, while other platforms serve for specialized applications. Table 5 summarizes these design tendencies. MuJoCo [161] is well aligned with $\mathcal{D}_1$ data, where stable local interaction and arm-centric rollouts matter most for robotic arm execution. Meta-World [193] introduced a widely used simulated benchmark for multi-task and meta-reinforcement learning across 50 robotic manipulation tasks. Later benchmarks such as LIBERO [106] and RoboCasa [126] added lifelong-learning curricula and richer household scenes. MJX and MuJoCo Playground support large-batch rollouts, differentiable simulation, and sim-to-real analysis within the same stack [196]. SAPIEN [182] is particularly suitable for $\mathcal{D}_2$ support, as articulated object states and part-level annotations support structured, temporally extended interaction modeling for robot manipulation. PartNet-Mobility provides articulated assets with annotated joints, ManiSkill builds benchmark suites on this basis [158], and SIMPLER uses the same infrastructure for sim-to-real evaluation [100]. NVIDIA Isaac is the platform best aligned with $\mathcal{D}_3$, particularly in settings that demand multimodality, scale, and broad platform coverage [122]. Representative embodied benchmarks built on NVIDIA Isaac Sim include BEHAVIOR-1K, which targets large-scale household activities, and ARNOLD, which focuses on language-grounded manipulation with continuous object states [47, 93]. Beyond these three ecosystems, several other platforms are still used for specialized applications of robotic arm manipulation. PyBullet and CoppeliaSim support manipulation baselines such as RLBench and CALVIN [72, 121]. Habitat and AI2-THOR help compensate for the limited semantic support available in mainstream manipulation simulators, supporting benchmarks such as ALFRED and OpenEQA [118, 151]. SoftGym supports deformable object manipulation benchmarks that rigid-body simulators still capture only weakly [105]. By reducing dependence on physical robot hardware, simulation improves experimental efficiency and allows more effort to be directed toward algorithmic questions above the arm hardware layer. However, simulation cannot fully capture real-world physics and hardware-specific behavior, and the target arm itself is not involved in the data collection loop. Consequently, the resulting $\mathcal{D}$ for robotic arm learning is often not fully faithful to robotic manipulation, leading to a substantial sim-to-real gap.

3.2.3 Automatic Data Generation. Building on the environments discussed above, Automatic data generation synthesizes different parts of $\mathcal{D}$, we discuss automatic data generation in three forms: scene generation synthesizes sensory observations, demonstration synthesis generates motion sequences together with contact events, and generative world models synthesize state transitions along with their future rollouts. Scene generation synthesizes sensory observations by widening the scene and semantic conditions under which they are produced within the observation space. ProcTHOR [38] generated large-scale procedural indoor environments, Holodeck [189] introduced semantic scene generation for embodied environments, and RoboGen [170] connected asset retrieval, scene construction, and policy training within one automated pipeline. Demonstration synthesis generates motion sequences together with contact events by generating additional robotic arm trajectories from limited source demonstrations or task specifications. MimicGen [120] showed that one demonstration can be expanded into many valid trajectories through spatial transformation, Tether [103] further broadens demonstration synthesis into autonomous trajectory generation, and GenSim [168] moved this idea to task generation by using LLMs to produce simulation code and reward structure. Generative world models synthesize state transitions along with their future rollouts by learning predictive models that can synthesize additional transitions from existing data. Cosmos [2] extends this paradigm to web-scale video data, training video foundation models to acquire physical priors from large-scale visual experience. IRASim [219] brings this into the real-robot setting through video world modeling of fine-grained robotic manipulation. DreamGen uses learned world models as both data generators and training environments [74]. Automatic data generation, however, synthesizes $\mathcal{D}$ by recombining or rolling forward an existing environment. The errors and limitations of that environment are also amplified.

3.3 Data Refinement

In real-world settings, available $\mathcal{D}$ is often still limited in scale and uneven in quality, which motivates further data refinement and augmentation. We organize these methods around four mechanisms: visual and viewpoint augmentation in Section 3.3.1 broadens sensory observations, physically grounded augmentation in Section 3.3.2 preserves the executability of motion sequences together with contact events under kinematic and contact constraints, correction and recovery augmentation in Section 3.3.3 supplements state transitions with the failure-recovery data, and data curation and verification in Section 3.3.4 determine whether existing $\mathcal{D}$ is still usable for learning and how low-quality data can be filtered, reweighted, or curated.

3.3.1 Visual and Viewpoint Augmentation. Visual and viewpoint augmentation expands existing demonstrations along two main axes: semantic appearance variation and camera transformation, both without requiring new robot rollouts. Early work such as GenAug and ROSIE used generative image models to vary textures, objects, and scene composition in order to widen the visual distribution seen during imitation learning [28, 194]. Later methods moved beyond appearance randomization to augmentations that explicitly model changes in camera viewpoint: RoVi-Aug augments both robot appearance and camera pose while preserving action labels and contact consistency under novel views [27]. A complementary line extends supervision beyond robot demonstrations: VISTA [159] enables multi-view synthesis from a single view for policy learning, H2R [94] converts egocentric human video into robot-centric pretraining data by replacing human hands with rendered end-effectors, and Pan et al. [131] combine novel view synthesis with trajectory warping to derive diverse training trajectories from a single demonstration.

3.3.2 Physically Grounded Augmentation. Augmenting physical trajectories (e.g., shifting a grasp 5 cm laterally) requires that the result still obey kinematic limits, collision constraints, and contact dynamics. CP-Gen [104] integrates kinematic boundaries with contact dynamics for one-shot trajectory augmentation. Yang et al. [187] combine VR human demonstrations, retargeting, and trajectory optimization to maintain dynamic feasibility across embodiments. CyberDemo [167] transfers simulated human demonstrations to real-world dexterous policies via large-scale augmentation in simulation, using minimal real data. Beyond geometric feasibility, RoCoDA uses counterfactual demonstration augmentation to preserve actions under causally irrelevant changes and transform actions consistently under rigid object transformations [5].

3.3.3 Correction and Recovery Augmentation. Most datasets lack failure-recovery examples, making policies fragile when they encounter Out-of-distribution (OOD) cases. DMD [204] uses diffusion models to synthesize corrective observations from perturbed states, enabling recovery without physical collection. Complementing these offline methods, IntervenGen [62] and CR-DAgger [185] systematize online human intervention during autonomous rollouts, so that sparse human corrections can be used to train policies to recover from mistakes.

3.3.4 Data Curation and Verification. As robot datasets scale in size and heterogeneity, the bottleneck shifts from data quantity to data utility. Existing methods differ primarily in how they estimate data value. ReMix [55] learns how to weight different dataset domains during pretraining so as to improve robustness across downstream domains. SCIZOR [207] prunes transitions that are weakly informative for task progress or redundant with other samples. CUPID [3] uses influence estimates to assess the contribution of each demonstration to a policy's closed-loop performance, and then uses these estimates for filtering and data selection.

3.4 Data Pipeline

In current practice, $\mathcal{D}$ is increasingly managed over a unified lifecycle, where collection, generation, augmentation, and refinement are treated as parts of one integrated production lifecycle. Existing systems reflect different ways of organizing this full lifecycle of $\mathcal{D}$. RoboTwin [125] keeps synthetic data tethered to measured physical setups by matching dual-arm kinematics and sensor noise, and its second version adds task synthesis for automatic scenario generation. RoboVerse [46] addresses fragmentation across simulators and datasets by providing a standardized MetaSim backbone spanning many tasks and assets. At larger scale, AgiBot World Colosseo [20] couples standardized large-scale real-robot collection with a generalist policy designed to better exploit heterogeneous data sources, while NVIDIA Isaac GR00T [128] packages synthetic data generation, collection, refinement, and sim-to-real alignment into a large stack built on Isaac infrastructure. Supporting these pipelines, AIRSPEED [181] decouples collection and generation from specific hardware backends, LeRobot [22] turns fragmented real-world trajectories into reusable repositories, and MimicLabs [140] studies how quantity, diversity, and quality trade off as pipelines scale. Taken together, these infrastructures move $\mathcal{D}$ production toward automated and reproducible workflows.

4 Learning Paradigms for Robotic Arm Systems

Learning paradigms are built on $\mathcal{D}$, introduced in Section 3.1, from which they acquire the knowledge support $\mathcal{K}$ required for robotic manipulation under embodiment constraints. In this sense, learning is the stage at which embodied intelligence becomes operational in robotic arm manipulation. Robotic arm manipulation can be organized into two primary learning paradigms according to how $\mathcal{K}$ is acquired and organized: cognitive planning, denoted by B, and policy learning, denoted by P. This distinction is reflected in the components of $\mathcal{D}$ that each paradigm primarily emphasizes. Cognitive planning (B) relies primarily on sensory observations and state transitions

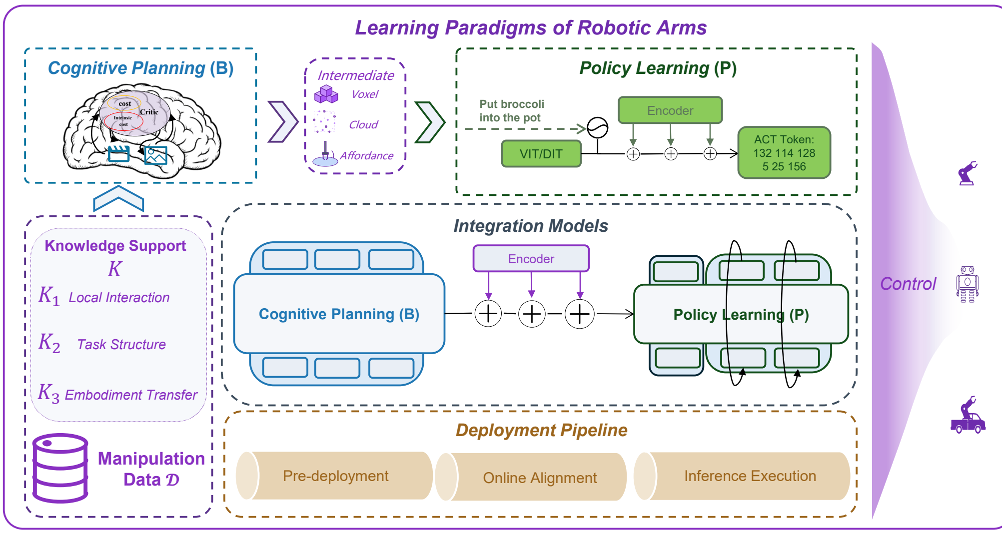

Fig. 4. Overview of learning paradigms for robotic arm manipulation. The figure organizes the pipeline into cognitive planning, policy learning, integration models, and deployment stages, linked by multimodal inputs and downstream control across robotic arm embodiments.

to construct intermediate artifacts and specify actions, whereas policy learning (P) relies primarily on contact events and motion sequences to produce executable motor behavior under different action representations. Integration models couple these two pathways into unified architectures, either explicitly or implicitly. As $\mathcal{D}$ expands from local interaction traces to task-level execution and cross-embodiment experience, the knowledge support $\mathcal{K}$ acquired from it also expands in scope. We distinguish three corresponding levels of $\mathcal{K}$: $\mathcal{K}_1$ supports learning local interaction, $\mathcal{K}_2$ supports learning task structure for execution, and $\mathcal{K}_3$ supports learning transferable structure across embodiments. Section 4.1 introduces cognitive planning paradigms, Section 4.2 covers policy learning paradigms, Section 4.3 examines how the two pathways are coupled, and Section 4.4 discusses deployment through a three-stage pipeline. Figure 4 summarizes this learning ecosystem.

4.1 Cognitive Planning: The Embodied Brain

Cognitive planning places greater emphasis on sensory observations and state transitions in $\mathcal{D}$, from which it acquires the support needed to construct intermediate artifacts. The primary differences among these methods lie in the intermediate artifacts through which that support is organized. We therefore classify cognitive planning into five paradigms (B1–B5) according to the primary intermediate artifact each paradigm produces. B1 specifies what should be executed in semantic form. B2 decomposes intent into logical task structure. B3 grounds intent in multimodal embodied context. B4 exposes spatial structure for manipulation. B5 predicts future interaction for planning and action evaluation. Among these paradigms, B1, B2, and B5 place greater emphasis on state transitions in $\mathcal{D}$, because their intermediate artifacts are organized around task progression and future evolution, whereas B3 and B4 place greater emphasis on sensory observations, because their intermediate artifacts are grounded in the current embodied scene. Meanwhile, the intermediate artifacts associated with B1–B5 collectively suggest a broader trend toward forms that are increasingly grounded and closer to execution. Table 6 summarizes these paradigms, while Figure 5 illustrates the overall organization within the learning paradigms.

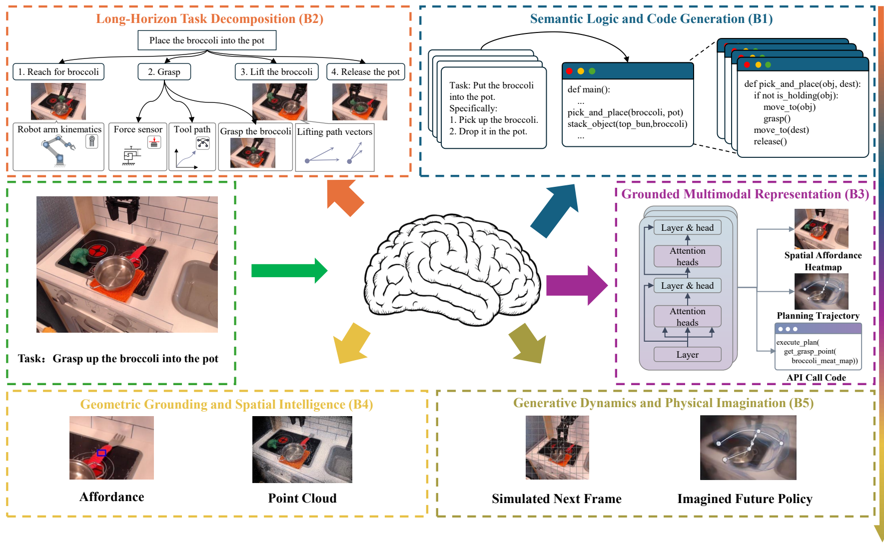

Fig. 5. Cognitive planning paradigms for embodied manipulation. The figure summarizes five planning modes discussed in this section.

4.1.1 B1: Semantic Logic and Code Generation. B1 methods translate high-level task intent into explicit specifications for execution, including skill selections, programs, reward functions, and behavior trees. Skill selection is the first specification type. SayCan [4] grounds abstract language goals by scoring primitive skills according to both semantic relevance and execution feasibility, but its output remains a ranked selection over predefined skills rather than a compositional execution program. Program synthesis provides a second form of specification. CaP [102] generates executable Python from natural language, while ProgPrompt [154] embeds precondition assertions into the prompt structure to enforce logical constraints. Demo2Code [169] extends this direction beyond language-only input by translating visual demonstrations into symbolic code. Reward synthesis provides a third form of specification. Tan et al. [195] synthesize programmatic reward functions that decouple semantic task specification from continuous trajectory optimization. Text2Touch [43] extends reward synthesis to tactile dexterous manipulation by using LLMs to design reward functions over tactile and proprioceptive signals. Structured control flow provides a fourth form of specification. LLM-BT [214] generates behavior trees that can be dynamically updated under environmental changes, while LLM-as-BT [7] generates modular behavior trees that support reactive switching and error recovery. Overall, B1 methods formulate task structure as structured semantic specifications. They mainly require $\mathcal{K}_2$, because their outputs are organized around task semantics, skill preconditions, objectives, and control logic rather than local sensorimotor dynamics.

4.1.2 B2: Long-Horizon Task Decomposition. B2 converts intent into logical task structures, decomposing long-horizon goals into ordered subtask sequences or formal plans. LLM+P [107] addresses this by translating natural language into PDDL and delegating reasoning to classical solvers that guarantee feasibility. L3M+P [1] further enables the solver to learn domain transition models from experience, reducing reliance on pre-defined rules. Classical solvers, however, face combinatorial explosions in large environments. SayPlan [136] mitigates this by anchoring search in 3D scene graphs, hierarchically pruning irrelevant regions before planning. Inner Monologue [68] extends this paradigm to replanning with feedback, while Statler [192] maintains a task-relevant world

Table 6. **Taxonomy of Cognitive Planning Paradigms.** ID denotes the paradigm identifier; Output Abstraction denotes the generated artifact; Execution Gap specifies what additional computation converts the output into motor commands.

ID	Support	Sub-direction	Output Abstraction	Execution Gap	Representative Methods
B1	$\mathcal{K}_2$	Skill Selection	Skill ranking	Skill API	SayCan [4]
		Program Synthesis	Executable code	Runtime interp.	CaP [102], Demo2Code [169]
		Reward Synthesis	Reward functions	RL optimizer	Language to Rewards [195]
		Structured Control Flow	BT / XML	Reactive exec.	LLM-BT [214], Ao et al. [7]
B2	$\mathcal{K}_2$	Formal Solvers	Symbolic task plan	Classical planner	LLM+P [107], L3M+P [1]
		Semantic Search	Sub-task sequence	Motion planner	SayPlan [136]
		State-Aware Planning	Re-planned task state	Skill API	Huang et al. [68], Statler [192]
B3	$\mathcal{K}_2/\mathcal{K}_3$	Multimodal Grounding	Grounded reasoning state	Skill API	PaLM-E [39], RoboBrain [76]
		Trace Grounding	Task trace / subgoals	Planner API	Mu et al. [124], ECoT [197]
B4	$\mathcal{K}_1/\mathcal{K}_2$	Physical Constraint	Contact / 6-DoF pose	Motion planner	ManipLLM [99], A3VLM [67]
		2D Affordance	Heatmap / mask	Grasp planner	CLIPort [152], PASG [220]
		3D Neural Fields	Neural field / 3D repr.	Motion planner	F3RM [148], D3Fields [171]
		Constraint Reasoning	Value map / keypts	Trajectory opt.	VoxPoser [69], ReKep [70]
B5	$\mathcal{K}_2/\mathcal{K}_3$	Future Observation Prediction	Future frames	Inverse dynamics	UniPi [40], RoboDreamer [215]
		World Action Prediction	Action dynamics	Policy planning	UWM [218], V-JEPA2 [8]
		Physics-Constrained Prediction	Trajectory / constraints	MPC / planner	Schulze et al. [142], [201], [127]

state throughout execution. Since they depend on the organization of long-horizon task structure and subtask relations, they primarily require $\mathcal{K}_2$ as well. The shared limitation of B1 and B2 is that semantic or logical correctness does not guarantee physical executability.

4.1.3 B3: Grounded Multimodal Representation. B3 grounds task intent in multimodal embodied context, producing intermediate representations that guide skill selection, task planning for downstream policy execution. PaLM-E [39] injects visual observations and robot state representations into the LLM embedding space, so that semantic reasoning is conditioned on the current embodied scene. EmbodiedGPT [124] incorporates current observations into step-by-step reasoning, making task progress more explicit during execution. ECoT [197] includes richer intermediate traces, including task descriptions, object grounding, and stepwise plans before action prediction. RoboBrain [76] integrates task reasoning with affordance, trajectory, and longer-horizon visual context for manipulation. Within B3, task-conditioned scene grounding mainly relies on $\mathcal{K}_2$, while more transferable multimodal structure draws on $\mathcal{K}_3$. The main limitation of B3 is that multimodal grounding may lack manipulation-specific physical commonsense, especially when tasks require reasoning about contact forces, support geometry, or irreversible state transitions during execution.

4.1.4 B4: Geometric Grounding and Spatial Intelligence. B4 outputs spatial representations, including contact regions, 6-DoF poses, affordance maps, and neural fields, that planners and controllers can consume directly. ManipLLM [99] predicts pixel-wise contact masks from language, and A3VLM [67] extends this grounding to articulated objects by inferring joint axes and motion types. For 2D affordance mapping, CLIPort [152] combines semantic grounding with spatial action localization for language-conditioned robotic arm manipulation, while PASG [220] strengthens the mapping with functional keypoints. For 3D affordance mapping, F3RM [148] injects open-vocabulary semantics into neural feature fields, while D3Fields [171] extends this line with dynamic 3D descriptor fields that encode semantic features and instance masks and support goal specification from 2D images. StructDiffusion [112] similarly maps high-level language goals and partial scene observations to physically valid target object arrangements for downstream execution. VoxPoser [69] and ReKep [70] push this line further by turning language or task descriptions into explicit 3D value maps and spatio-temporal relational keypoint constraints for downstream optimization. B4 methods primarily represent the current geometric state of interaction, and therefore mainly require $\mathcal{K}_1$. When extended toward high-level constraint reasoning, they increasingly require $\mathcal{K}_2$. The main limitation of B4 is that spatial representations mainly specify where and how to act in the current scene, but provide limited support for reasoning about how contact changes object state over time.

4.1.5 *B5: Generative Dynamics and Physical Imagination.*

B5 focuses on predictive planning artifacts that support reasoning about future interaction. Future Observation Prediction generates future visual rollouts that can be converted into actions through inverse dynamics or downstream action selection. UniPi [40] reformulates policy as future video generation, while RoboDreamer [215] uses imagined visual rollouts to support planning. World Action Prediction models future evolution in a more abstract dynamics space, where the resulting predictions can support policy decoding or planning. UWM [218] unifies video and action diffusion within one transformer, making future actions part of the generative dynamics, while V-JEPA2 [8] shifts from pixel reconstruction to latent predictive reasoning. Physics-constrained prediction makes imagined rollouts more compatible with planners or MPC-style controllers by restricting future trajectories with structural priors. Schulze et al. [142] inject analytical mechanics into learned dynamics, Zhang et al. [201] targets deformable interaction through particle-mesh structure, and LUMOS [127] uses language instructions to guide predictive rollouts in a learned world model. Within B5, task-specific predictive rollouts mainly rely on $\mathcal{K}_2$, while world-action models draw on $\mathcal{K}_3$ when they learn transferable latent dynamics across tasks or embodiments. The main limitation of B5 is that errors in imagined rollouts can accumulate, especially under contact-rich dynamics and long-horizon execution.

4.2 Policy Learning: The Embodied Cerebellum

Policy learning maps perceptual observations to motor commands, thus focusing more on the contact events and motion sequences in $\mathcal{D}$. We organize these methods into three categories by how the action space is represented. Explicit Continuous Policies (P1) output continuous action vectors via regression in a single forward pass. Discrete Action Policies (P2) quantize actions into tokens and generate them autoregressively via cross-entropy. Iterative Generative Policies (P3) model the full action distribution through multi-step denoising or flow matching. Within these three categories, sub-directions further distinguish how policy generation is factorized in practice, how it handles distribution shift, and how it generalizes across embodiments. Table 7 summarizes overview of this kind of paradigms, and Figure 6 contrasts these paradigms.

4.2.1 *P1: Explicit Continuous Policies.*

P1 policies learn explicit mappings from observations or task conditions to continuous actions. Some methods rely on compact visual or geometric representations to support direct action mapping. BC-Z [73] uses this pattern for multi-task zero-shot generalization, MVP [135] shows that masked visual pre-training yields representations robust enough to reduce the policy to a simple MLP, Iterative Residual Policy [31] predicts continuous action corrections from previous executions, and PointPolicy [53] extends explicit regression to 3D point clouds. Other methods predict continuous action chunks over a short execution horizon. ACT-style methods [45, 209] combine action chunking with CVAE-based modeling to predict short-horizon manipulation sequences and reduce jitter in bimanual and mobile manipulation. BAKU [52] extends action chunking to multi-task manipulation with multimodal observations and transformer-based continuous action prediction. Some methods improve continuous action mapping under limited robot data by introducing auxiliary human data or motion representations. MT-ACT [17] improves data efficiency by combining semantic augmentations with action chunking, while MimicPlay [165] leverages human play data to guide low-level visuomotor control

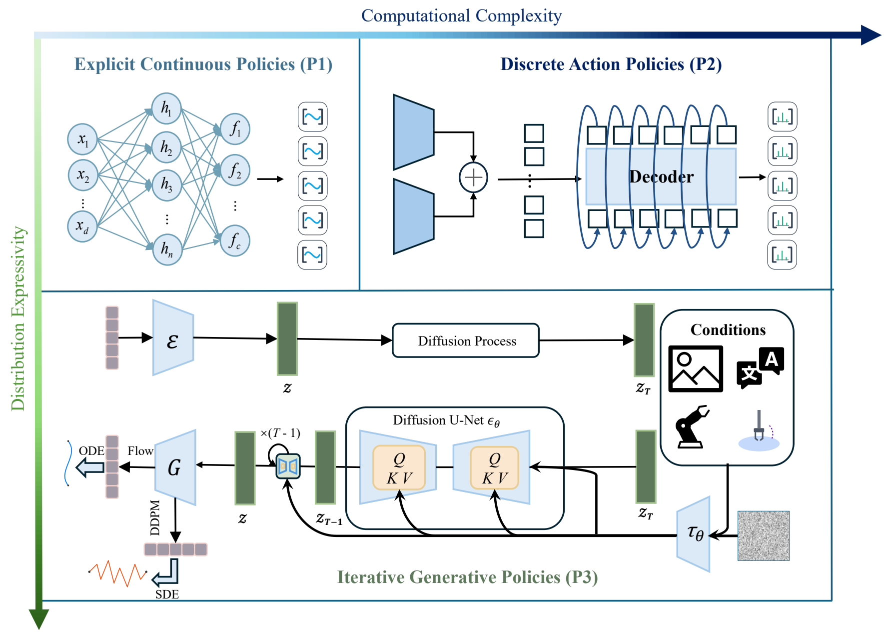

Fig. 6. Comparison of policy learning paradigms for robotic manipulation. The figure contrasts these paradigms in terms of distributional expressivity and computational complexity.

through hierarchical policy learning. and Ren et al. [137] learns a unified motion representation for cross-embodiment adaptation. Within P1, direct reactive policies mainly rely on $\mathcal{K}_1$, while task-conditioned and chunked variants draw more strongly on $\mathcal{K}_2$ through short-horizon execution structure. P1 offers efficient and deployable continuous control, but its explicit mapping form generally has weaker expressiveness for structured or multimodal action choices.

4.2.2 P2: Discrete Action Policies. P2 policies discretize continuous actions into tokens and model them autoregressively, enabling multimodal action generation beyond unimodal regression. Early approaches use predefined token spaces to discretize continuous actions. RT-1 [19] maps continuous action dimensions to to fixed bins, and PerAct [153] formulates 3D manipulation as action classification over voxelized spatial grids. However, fixed discretization imposes a rigid token space that may not match the structure of continuous action distributions. This motivates learned action tokenization, which adapts discrete action representations to demonstrated behavior. BeT [144] learns discrete action modes through k-means clustering and predicts continuous offsets, while VQ-BeT [91] uses vector quantization to learn a compact codebook of latent action tokens. Action tokenization also brings robot behavior into the autoregressive language-style sequence modeling framework. VIMA [78] interleaves visual objects with text instructions for multimodal prompt conditioning, while GR-1 [177] jointly predicts actions and future frames to leverage large-scale video pre-training. ICRT [44] treats imitation as in-context sequence prediction, enabling zero-shot adaptation to unseen tasks. Beyond context conditioning, recent work further explores the structure of action sequences in autoregressive generation. CARP [48] uses multi-scale action tokenization for coarse-to-fine autoregressive prediction. Seer [160] predicts future visual states as an intermediate goal and derives actions via inverse dynamics. ARP [205] organizes actions into chunks for autoregressive generation, achieving robust performance across diverse manipulation tasks.

Table 7. **Taxonomy of Policy Learning Paradigms for Embodied Manipulation.** Scale indicates the dominant model-size tendency of representative policy models. Abbrev.: AR = Autoregressive; Cont. = Continuous.

ID	Support	Sub-direction	Action Space	Scale	Input	Modeling	Methods
P1	$\mathcal{K}_1/\mathcal{K}_2$	Direct Mapping	Cont.	<100M	2D / 3D	Direct regression	BC-Z [73], MVP [135]
		Action Chunking	Cont. chunk	<100M	2D / multi-view	Chunked action	ACT [209], BAKU [52]
		Enhanced Mapping	Cont.	Mixed	2D / video	Goal-conditioned	MT-ACT [17], [165], [137]
P2	$\mathcal{K}_2/\mathcal{K}_3$	Fixed Discretization	Fixed bins	<100M	2D / 3D	Discr. categorical	RT-1 [19], Peract [153]
		Learned Discretization	Codebook	<100M	2D	Latent discrete	BeT [144], VQ-BeT [91]
		Action Token AR	MM tokens	Mixed	2D / multi-modal	Prompt-conditioned	VIMA [78], [177], [44]
		Sequence AR	Sequence tokens	<100M	2D / multi-modal	Autoregr. sequence	Seer [160], [205], [48]
P3	$\mathcal{K}_1/\mathcal{K}_3$	Diffusion Generative	Denoised cont.	100–500M	2D	Diffusion denoising	Diffuser [75], DP [33]
		Generative 3D	Denoised cont.	<100M	3D	Geometry-aware	DP3 [198], iDP3 [199]
		Flow-based Generative	Flow cont.	<100M	2D / 3D	Flow Matching	Jiang et al. [77], [202]
		Generative Scaling	Denoised cont.	Mixed	2D / multi-view	Scaled generative	Octo [129], [111], [138]

Within P2, task-level token policies mainly rely on $\mathcal{K}_2$, since discrete action tokens are organized around task execution structure. Multimodal and generalist token policies draw on $\mathcal{K}_3$ when their token spaces support transferable action structure across heterogeneous embodiments. The central challenge is to align discrete action tokens with the structured and continuous nature of robot actions, while preserving action fidelity and efficient autoregressive generation.

4.2.3 P3: Iterative Generative Policies. P3 policies treat action prediction as conditional generative modeling over continuous action trajectories, progressively refining noisy trajectory samples into executable action sequences to model multimodal action distributions. Early diffusion-based policies first appeared in planning and structured generation. Diffuser [75] casts long-horizon trajectory generation as probabilistic inpainting. Diffusion Policy (DP) [33] subsequently brings this paradigm into mainstream visuomotor control by learning the score function of the action distribution, thereby avoiding the mode-averaging limitations of direct regression. Consistency Policy [133] further distills diffusion-based action generation into few-step consistency generation. In 3D robotic arm manipulation, DP3 [198] further conditions diffusion policy on point-cloud observations for spatial grounding, while iDP3 [199] improves the underlying 3D representations for more robust deployment. Despite these advances, diffusion generative methods inherit the resource cost of multi-step denoising. Flow matching addresses this by learning velocity fields that transport noise to data along straighter trajectories with fewer integration steps. FlowPolicy [202] improves flow-based policy generation through consistency flow matching in 3D robotic manipulation, and Streaming Flow Policy [77] streams action trajectories to the robot during generation for online execution. This generative action-modeling approach has also been scaled toward generalist robot policy learning. Octo [129] trains a transformer-based diffusion policy on OXE data as a general-purpose policy initialization. RDT-1B [111] extends diffusion-based action generation to heterogeneous multi-robot data by learning a shared action representation for complex manipulation. FLOWER [138] further improves sample efficiency through flow-matching. Within P3, local generative control mainly relies on $\mathcal{K}_1$, while generalist generative policies draw on $\mathcal{K}_3$ when they capture transferable action regularities across embodiments. Overall, P3 offers strong multimodal action modeling, but deployment requires faster generation without degrading action quality or control stability.

4.3 Integration Models

To learn from the full content of $\mathcal{D}$, robotic arm manipulation systems start to increasingly integrate B with P. RT-2 [221] marked an early demonstration of integration models, showing that semantic

Table 8. **Integration patterns across data regimes.** The table summarizes recurrent integration patterns for robotic arm manipulation in terms of architecture type, supporting data regime, coupling principle, shared state, and typical pairings of cognitive planning and policy learning paradigms.

Pattern	Type	Support	Coupling Principle	Shared State	Typical B	Typical P
Explicit Hierarchy	II	$\mathcal{K}_2$	Explicitly decoupled	Language / code / task trace	B1 / B2	P1 / P2
Unified End-to-End	I	$\mathcal{K}_3$	Shared backbone	Minimal / implicit	B3	P2 / P3
Implicit Hierarchy	II	$\mathcal{K}_2/\mathcal{K}_3$	Implicitly decoupled	Latent / token / soft split	B3 / B2	P2 / P3
Spatial Grounding	III	$\mathcal{K}_1/\mathcal{K}_3$	Shared explicit geometric	Maintained spatial state	B4	P1 / P2 / P3
Predictive WM	III	$\mathcal{K}_2/\mathcal{K}_3$	Shared explicit predictive	Maintained predictive state	B5	P3

understanding, reasoning, and action generation can be jointly realized within a shared policy model. Later work has evolved into five integration patterns, which can be grouped into three coupling types. Type I emphasizes end-to-end unification within a shared backbone. Type II covers hierarchical designs that separate slower reasoning from faster execution. Type III maintains an explicit spatial or predictive world representation to mediate action generation. Table 8 summarizes representative methods, and Table 9 further lists representative integrated paradigms.

4.3.1 Type I: Unified End-to-End Foundation Models. Type I models integrate perception, instruction-conditioned grounding, and action generation within a shared backbone. Early Type I models often adopted discrete action (P2) heads for action generation, as it provides a straightforward way to represent robotic arm manipulation within a unified integration model. RT-2 [221] co-fine-tuned a VLM to output discrete action tokens, and OpenVLA [88] released a 7B model trained on 970k OXE trajectories. X-VLA [211] continues this line with embodiment-specific soft prompts inside a shared transformer, showing how a unified backbone can extend across multiple robot platforms without introducing an explicit hierarchical split. UniVLA [172] further unifies vision, language, action, and video prediction as discrete token sequences within a shared autoregressive framework. More recent Type I models further incorporate iterative generation (P3) head for continuous action prediction. π_0 [18] integrates a pretrained vision-language backbone with flow-matching action generation within a unified backbone, producing high-frequency continuous actions across diverse manipulation tasks. RoboMamba [109] replaces self-attention with state space models for linear-time processing. ReconVLA [156] adds reconstruction-based auxiliary objectives to strengthen 3D spatial encoding within the unified backbone. Although Type I models demonstrate strong benefits from unifying perception, semantic reasoning, and action generation within a single architecture, this unification also comes at a cost. Because semantic grounding and action generation are integrated into the same unified backbone, the model needs to process heterogeneous information, which makes joint representation learning more challenging.

4.3.2 Type II: Hierarchical & Modular Reasoning Models. Type II models separate slower reasoning from faster execution through hierarchical or modular interfaces. Within Type II, explicit models expose intermediate artifacts, whereas implicit models represent them as latent internal representations. Representative explicit Type II methods include RT-H [16] introduced language motions as an intermediate layer between high-level instructions and low-level actions, CoT-VLA [208] and ThinkAct [66] expose intermediate reasoning steps, and HAMSTER [101] predicts a coarse 2D path as an intermediate representation to guide a low-level policy for precise 3D manipulation. Representative implicit Type II methods include FiS-VLA [25] couples a slower semantic branch with a real-time execution branch, GO-1 [20] similarly uses a latent planner to bridge multimodal

Table 9. **Representative Methods of Integration Models for Robotic Arm Manipulation.** Methods are summarized by their coupled B/P components, integration mode, hardware setting, and data form. **Mode**: Dir. = Direct Control; Reas. = Reasoned Control; Hier. = Hierarchical Control; Genr. = Generative Control. **Hardware**: Div. = Diverse multi-robot; Prop. = Proprietary. **Data Form**: Traj. = trajectories; Real Traj. = real robot trajectories; Syn. = synthetic; Web Video = off-domain web videos.

Model	*B & P*	Backbone & Params	Mode	Hardware	Primary Data Source	Eval: Sim & Real
Type I: Unified End-to-End Foundation Models						
RT-2 [221]	B1, P2	PaLI-X (55B)	Dir.	Google Robot	130K Real Traj.	Real: 700 manipulation tasks
OpenVLA [88]	B3, P2	Llama-2 + SigLIP (7B)	Dir.	Franka+WidowX	970K OXE Real Traj.	Real: 29 manipulation tasks
π**0** [18]	B3, P3	PaliGemma (3B)	Genr.	Div. (8 types)	~10K OXE Real Traj.	Real: 7 dexterous tasks
RoboMamba [109]	B3, P3	Mamba SSM (3B)	Dir.	Franka	OXE Real Traj.	Sim: CALVIN | Real: manipulation tasks
ReconVLA [156]	B4, P3	LLaVA (7B)	Genr.	AgileX PiPer	2M samples (100K+ Traj. BridgeV2/LIBERO)	Sim: CALVIN (ABCD→D) | Real: 5 manipulation tasks
UniVLA [172]	B5, P2	Emu3 (8.5B)	Dir.	AgileX	OXE Real Traj.	Sim: LIBERO (95.5%) | Real: 8 ALOHA tasks
X-VLA [211]	B3, P2	Transformer (0.9B)	Genr.	Div. (3 types)	290K episodes	Sim: LIBERO 93%, Simpler-WidowX | Real: 3 robots
Type II: Hierarchical & Modular Reasoning Models						
RT-H [16]	B2, P2	PaLI-X (7B+)	Hier.	Google Robot	Kitchen Real Traj.	Real: 8 manipulation tasks
CoT-VLA [208]	B2, P2	VILA-U (7B)	Reas.	Franka+WidowX	OXE Real Traj. + EPIC	Real: 10 long-horizon tasks
ThinkAct [66]	B1, P2	Qwen2.5-VL (7B)	Hier.	Franka	OXE + RoboVQA Real Traj.	Sim: CALVIN | Real: 10 long-horizon tasks
FiS-VLA [25]	B2, P1	Llama-2 (7B)	Hier.	Agilex	860K real trajectories	Sim: RLBench | Real: 8 manipulation tasks
HAMSTER [101]	B4, P1	VILA-1.5 (13B)	Hier.	Franka	Web Video	Sim: Colosseum | Real: 74 tasks / 222 evals
Hi Robot [150]	B2, P3	PaliGemma (3B)	Hier.	Div. (3 types)	Syn. Data	Real: 3 open-ended domains
GO-1 [20]	B3, P3	InternVL (2B)	Hier.	Div.	1M+ Real Traj. + Web Video	Sim: LIBERO / GenieSim | Real: 5 tasks
Type III: Spatial Grounding & Predictive World Models						
3D-VLA [210]	B4, P3	Flan-T5-XL (3B)	Genr.	Franka	2M samples OXE (3D)	Sim: RLBench | Real: 10 3D tasks
VideoVLA [149]	B5, P3	CogVideoX (5B)	Genr.	Realman	OXE + 5K samples	Sim: SIMPLER | Real: 6 unseen tasks
SpatialVLA [134]	B4, P2	PaLiGemma 2 (4B)	Dir.	Franka+WidowX	OXE + RH20T Real Traj.	Sim: SIMPLER / LIBERO | Real: 24 real-robot tasks
TraceVLA [212]	B4, P3	Llama-2 + SigLIP (7B)	Dir.	WidowX-250	OXE + 150K Real Traj.	Sim: SIMPLER (137 configs) | Real: 4 WidowX tasks
DreamVLA [203]	B5, P3	Diffusion WM (7B)	Genr.	Franka	CALVIN Real Traj.	Sim: CALVIN | Real: 10 manipulation tasks
BridgeVLA [97]	B4, P2	PaliGemma (3B)	Dir.	Franka	Bridge / OXE Real Traj.	Sim: RLBench/COLOSSEUM | Real: 13 Franka FR3 tasks
DreamZero [191]	B5, P3	Wan2.1 (14B)	Genr.	Div. (3 types)	Web Video + Real Traj.	Sim: DROID/PolaRiS/Genie | Real: 60 manipulation tasks

understanding and action generation through latent action tokens, and Hi Robot [150] adopts a hierarchical paradigm design for open-ended instruction following and user feedback during execution. RoboInter [95] broadens Type II integration by turning intermediate representations into supervision and evaluation targets. Although explicit and implicit Type II models offer advantages, their drawbacks are also evident. Explicit interfaces are easier to inspect and correct, but they may lose execution-sensitive information when task structure is compressed into fixed intermediate representations. Implicit hierarchies, by contrast, preserve a flexible internal interface, but make it harder to localize failures across reasoning, grounding, and execution. In consequence, separating

Table 10. **Deployment bottlenecks and representative techniques across the 3-Stage pipeline.** **S1** = Offline Pre-deployment; **S2** = Online Physical Alignment; **S3** = Inference-Time Execution.

Stage	Deployment Bottleneck	Technique Category	Representative Methods
S1	Offline RL at Scale	Autoregressive Q-fn, offline TD backups	Q-Transformer [24], TRANSIC [79]
	Cross-Embodiment	R2S2R latent action spaces	X-Sim [36]
	Sim-to-Real & Compression	Delta action compensation, DCT tokenization	ASAP [54], FAST [132]
S2	RL Fine-Tuning	Direct GRPO, world-model-guided adaptation	SimpleVLA-RL [96], DiWA [23]
	Contact-Rich Adapt.	Consistency policies, visuo-tactile refinement	ConRFT [30], VT-Refine [65]
	Forgetting & Residual	Off-policy residual RL	PLD [183], OpenVLA-OFT [87]
S3	Quantization	Action-centric mixed-precision, 1-bit PTQ	QVLA [186]
	Decoding Acceleration	Parallel fixed-point iterations	TinyVLA [176], PD-VLA [155]
	Robustness & Safety	Contrastive decoding, motion planning	Policy CD [179], Neural MP [35]
	Failure Detection & Recovery	Flow-based OOD detection, runtime updates	FAIL-Detect [184], Reflective TTP [61]

slow planning from fast control can induce information loss by forcing intermediate representations to align across heterogeneous abstraction between high-level reasoning and low-level control.

4.3.3 Type III: Spatial Grounding & Predictive World Models. Type III models mediate action generation through explicit world representations. Within Type III, spatial grounding models expose geometric state for action generation, whereas predictive world models represent future interaction by modeling state transitions over time. Representative spatial grounding methods include 3D-VLA [210], which embeds 3D perception into the LLM for spatial reasoning, SpatialVLA [134], which grounds actions in metric 3D structure via Ego3D Position Encoding, TraceVLA [212], which conditions generation on visual trajectory traces, and BridgeVLA [97], which aligns perception and action prediction through spatial heatmap representations to improve kinematic feasibility. Representative predictive world models include VideoVLA [149], which jointly predicts manipulation sequences and future visual outcomes using a pre-trained video generator, DreamVLA [203] which uses a diffusion-based world model to support generative planning through imagined trajectories, and DreamZero [191], which learns a joint predictive model over future visual states and actions. Although Type III models provide explicit world structure for grounding and prediction, their performance depends on the reliability of the shared representation. If this representation is inaccurate or stale, the system may produce physically misaligned actions. Correcting such representations often incurs substantial computational and engineering cost.

4.4 Deployment Pipeline

Reliable real-world execution requires learned manipulation policies to be adapted to the embodiment constraints of the target robotic arm. In practice, this adaptation is increasingly organized into a deployment pipeline. Before deployment, models are refined and prepared for physical rollout through offline adaptation, transfer, and compression. During deployment, they are further aligned to the target hardware and interaction conditions. At runtime, they are executed under real-world constraints with support for efficient, robust, and safe control. Table 10 summarizes the main bottlenecks and representative techniques at each stage.

4.4.1 Stage 1: Offline Pre-deployment. Before physical deployment, the main challenges are transferring policies from simulation to reality, generalizing across embodiments, and making large models practical for real-world execution. Q-Transformer [24] shows that autoregressive Q-functions can scale offline actor-critic learning to transformer policies through temporal difference backups on static datasets. X-Sim [36] uses a real-to-sim-to-real pipeline to learn cross-embodiment manipulation from human videos by treating object motion as a dense and transferable signal. ASAP [54] mitigates dynamics mismatch between simulation and reality by learning a residual action compensation model that aligns simulated dynamics with real-world execution. TRANSIC [79] adopts a complementary human-in-the-loop strategy in which operators provide online corrections during real-world execution, from which a residual policy is learned and combined with the original simulation policy for autonomous deployment. As model size grows, pre-deployment compression also becomes necessary: FAST [132] improves deployment efficiency by replacing naive action discretization with a compressed action representation based on the discrete cosine transform.

4.4.2 Stage 2: Online Physical Alignment. Once deployed on physical hardware, the primary bottleneck shifts from transfer preparation to adaptation under real contact, friction, sensing noise, and OOD objects. SimpleVLA-RL [96] shows that large manipulation models can be fine-tuned directly through reinforcement learning, FLaRe [64] addresses the tension between generalist priors and specialist performance through adaptive large-scale RL fine-tuning, while DiWA [23] adapts diffusion policies through an offline world model, reducing the amount of physical interaction needed for post-training improvement. ConRFT [30] combines offline BC and Q-learning with online consistency policies and human intervention checkpoints to support safe adaptation in contact-rich settings. Human-in-the-loop real-world reinforcement learning provides another effective route for this stage: HIL-SERL [115] combines demonstrations, learned reward modeling, and online human corrections to learn precise and dexterous manipulation skills directly on physical robots. VT-Refine [65] further demonstrates that visuo-tactile policies, when paired with tactile digital twins and refinement in simulation, can improve high-precision dual-arm coordination. Other alignment strategies include residual adaptation via off-policy RL [183]. The adaptation choices made here directly affect Stage 3: aggressive full-parameter fine-tuning can improve task performance but complicate compression and serving, whereas residual or modular updates preserve more of the original policy structure at the cost of additional execution-time overhead.

4.4.3 Stage 3: Inference-Time Execution. During real-world deployment, the key challenge is to sustain reliable closed-loop execution on edge hardware. QVLA [186] addresses action-sensitive quantization through channel-wise bit allocation, assigning higher precision to channels that are more sensitive in the action space. TinyVLA [176] shows that compact paradigm design can improve both inference speed and data efficiency, even eliminating the need for a separate large-scale pretraining stage, while PD-VLA [155] accelerates chunked action generation by replacing sequential autoregressive decoding with parallel decoding iterations. Runtime robustness is equally critical. Policy Contrastive Decoding [179] mitigates sensitivity to irrelevant visual features during action generation, FAIL-Detect [184] reframes runtime failure detection as sequential OOD detection without requiring failure data, and Neural MP [35] improves execution reliability by distilling classical motion planning into a fast reactive neural planner that generalizes across cluttered real-world settings. Reflective TTP [61] supports online recovery through reflection and updates during test, re-evaluating actions with hindsight to adapt the policy during deployment.

Despite progress across all three stages, current methods still face several limitations. Pre-deployment approaches are often sensitive to simulator fidelity, offline data coverage, and the scope of embodiment variation, which limits the robustness of transfer to open-ended real-world settings. Online physical alignment methods improve adaptation under real dynamics, but they often require expensive robot interaction, careful reward or intervention design, and, in many cases, substantial human involvement that may hinder scalability. At inference time, compression, quantization, and decoding acceleration improve deployability, yet their impact on fine-grained action fidelity and contact-rich control are insufficiently understood.

Table 11. **Evaluation protocols and representative works for robotic arm systems.**

Dimension	Metric Category	Representative Metrics	Representative Works
Execution Quality	Task Completion Trajectory Smoothness Dual-Arm Coordination	Stage Success, Task Progression SPARC, LDLJ, Path Length, Jerk Velocity Divergence, Height Discrepancy	RoboEval [173], [143] SPARC [12], LDLJ [6], [173] RoboEval [173]
Data Efficiency & Quality	Info-theoretic Gradient-based Sample Efficiency	Mutual Information, Demo Contribution Influence Scores, Hessian-based Valuation AULC	DemInf [56] CUPID [3], QoQ [90] General IL benchmarks
Generalization & Transfer	Human Preference OOD Generalization Sim-to-Real & World Models	Pairwise Comparison, VLM-based Eval ΔSR_{OOD}, RMSD MMRV, VLM-as-Judge, ΔSR	RoboArena [9], [217] REALM [143] SIMPLER [100], [50], [146]
Deployment Efficiency	Energy Decomposition Normalized Efficiency Multi-objective Optimization	$P_{\text{elec}}, P_{\text{friction}}, P_{\text{copper}}, P_{\text{mech}}$ $\eta_E(\mathbf{q})$, $f_U/f_C/f_R$, CoT Execution Time, Jerk, Energy	ECDP [58] Vidussi et al. [162], [59] Rezali et al. [139]

5 Evaluation Protocols for Robotic Arm Systems

Previous sections have examined how robotic arm systems are built by embodied foundations, data ecosystems, and learning paradigms. This chapter reviews current efforts to evaluate robotic arm systems from four corresponding perspectives. Section 5.1 evaluates whether cognitive planning (B) and policy learning (P) can organize the knowledge support $\mathcal{K}$ induced by $\mathcal{D}$ into successful task execution. Section 5.2 evaluates whether $\mathcal{D}$ provides data that is sufficiently efficient and informative to induce useful knowledge support $\mathcal{K}$. Section 5.3 evaluates whether the knowledge support $\mathcal{K}$ learned from $\mathcal{D}$, and organized through B and P, is robust beyond the training distribution under varying embodiment constraints. Section 5.4 evaluates whether the physically grounded capabilities produced by B and P from $\mathcal{K}$ can be executed efficiently on physical embodiments within the deployment pipeline. Table 11 summarizes representative evaluation protocols for robotic arm systems.

5.1 Task Execution Quality Evaluation

Task execution is a widely adopted evaluation criterion because it provides a direct measure of model success on the task. However, binary success rates are easy to interpret, but they cannot distinguish early failure from almost successful and unstable execution, especially in long-horizon or bimanual settings. Finer-grained metrics can be grouped into three categories. Task progression metrics are more closely aligned with $\mathcal{K}_2$-level task organization, trajectory quality metrics probe $\mathcal{K}_1$-level execution quality, and bimanual coordination metrics reflect execution coupling. For task completion, RoboEval [173] decomposes manipulation into sequential stages and measures progress as the fraction completed, REALM [143] further evaluates robustness under perturbations using the Root Mean Squared Deviation of task progression relative to the default setting:

$$\mathrm{RMSD}(p) = \sqrt{\frac{1}{MT}\sum_{m=1}^{M}\sum_{t=1}^{T}\left(r_{m,t}^{p} - r_{m,t}^{\text{default}}\right)^2}, \tag{1}$$

where $r_{m,t}$ denote the task progression for trial P under perturbed and default conditions. For trajectory quality, smoothness metrics such as SPARC [12] and LDLJ [6] quantify jerk and motion regularity, while Dynamic Time Warping measures alignment with expert demonstrations under temporal misalignment. For bimanual manipulation, RoboEval also measures coordination through velocity divergence and height discrepancy between end-effectors.

5.2 Data Efficiency and Quality Evaluation

Data efficiency and quality concern whether $\mathcal{D}$ is sufficiently informative and efficient to induce useful $\mathcal{K}$. Existing evaluation efforts in this direction are typically conducted along two dimensions. One dimension is whether a dataset provides broad coverage enough for knowledge induction. DemInf [56] estimates the contribution of each demonstration to the mutual information between states and actions, using this quantity to capture both state diversity and action predictability. Another dimension is whether demonstrations are useful for downstream learning. CUPID [3] estimates how individual demonstrations affect downstream policy performance, providing a way to rank training data by utility. QoQ [90] instead evaluates samples by their contribution to validation improvement, helping identify low-quality or detrimental demonstrations. Beyond these criteria, sample efficiency is commonly summarized by the Area Under the Learning Curve (AULC):

$$\mathrm{AULC} = \frac{1}{N_{\max}} \int_0^{N_{\max}} SR(n)\, dn \approx \frac{1}{N_{\max}} \sum_k SR(n_k)\, \Delta n_k \tag{2}$$

where $SR(n)$ is the success rate after training on n demonstrations. Together with coverage and utility estimates, it evaluates $\mathcal{D}$ beyond dataset size, while remaining dependent on the task distribution, embodiment, and learning algorithm.

5.3 Generalization and Transfer Evaluation

Generalization and transfer evaluation concerns whether the knowledge support $\mathcal{K}$ learned from $\mathcal{D}$ and organized through B and P is valid under distribution shift. This requires evaluating whether learned knowledge remains reliable across task, environment, and embodiment variations. Existing efforts in this direction can be grouped into three categories: human judgment, robustness under distribution shift, and proxy validity. Human judgment evaluates aspects of manipulation quality that are not fully reflected by automated success metrics. RoboArena [9] uses pairwise human comparison to infer global policy rankings across distributed real-world evaluations, while AutoEval [217] reduces human evaluation overhead by automating success detection and scene resets, thereby minimizing the need for manual trial-by-trial assessment. Robustness under distribution shift measures whether performance is stable outside the training distribution. REALM [143] evaluates generalization under visual, semantic, and behavioral perturbations, using changes in task progression and the Root Mean Squared Deviation of task progression to quantify degradation under shift, and further reporting binary success rate and time to completion at the task level. Proxy validity evaluates whether scalable testbeds faithfully reflect real-world policy performance. SIMPLER [100] measures this through Mean Maximum Rank Violation (MMRV):

$$\mathrm{MMRV}(R, R_S) = \frac{1}{N} \sum_{i=1}^{N} \max_{1 \le j \le N} \left(|R_i - R_j| \, \mathbf{1}\left[(R_{S,i} < R_{S,j}) \neq (R_i < R_j) \right] \right) \tag{3}$$

where R_i and $R_{S,i}$ denote the real-world and simulated performance of policy i, respectively. WorldArena [146] extends this idea to learned world models by asking whether they can serve as reliable evaluators in addition to simulators. Together, these methods range from real-world evaluation to more scalable settings, including simulators and learned world models.

5.4 Deployment Efficiency Readiness Evaluation

Deployment efficiency readiness concerns whether the physically grounded capabilities produced by B and P from $\mathcal{K}$ can be executed efficiently on physical embodiments under real deployment constraints. Relevant costs include latency, compute, memory, hardware load, and energy consumption. Existing evaluation efforts in this direction can be grouped into three categories: energy decomposition, normalized efficiency comparison, and multi-objective trade-off analysis. Existing energy evaluations usually begin with simplified power models or their proxies before moving to more structured decompositions. Another structured view is provided by ECDP [58], which decomposes total power into electronics, friction, copper loss, and useful mechanical work:

$$P_{\text{total}}(t) = P_{\text{elec}} + P_{\text{friction}}(t) + P_{\text{copper}}(t) + P_{\text{mech}}(t), \tag{4}$$

where P_{elec} denotes baseline electronic power, $P_{\text{friction}}(t)$ denotes frictional loss, $P_{\text{copper}}(t)$ denotes resistive motor loss, and $P_{\text{mech}}(t)$ denotes useful mechanical output. This decomposition makes energy evaluation more informative than a single aggregate power value. For comparison across robots, recent work introduces normalized efficiency metrics [59, 162] and measures such as Cost of Transport, while Rezali et al. [139] formulate trajectory planning as a multi-objective optimization problem over several metrics. Current robotic arm manipulation benchmarks still provide only limited visibility into deployment efficiency, since latency, runtime reliability, compute footprint, and hardware load are rarely reported in comparable ways.

Overall, existing evaluation efforts remain insufficient to close the validation loop across the full manipulation pipeline. Current evaluations rarely systematically trace task-specific failures back to the underlying bottlenecks in data, representation, embodiment, planning, or control.

6 Open Challenges

Building on the foregoing organizational framework, we examine the open challenges of robotic arm manipulation from the perspective developed above. Limitations originating in embodied foundations accumulate in data support, intensify in architectural design, and ultimately make evaluation an unstable source of feedback for robotic arm manipulation. We then examine trustworthiness when robotic arm manipulation systems are deployed in real-world settings. For each challenge, we also identify recent efforts that point toward future directions.

6.1 Embodiment Infrastructure with Reduced Constraints

Current embodied foundations are confined to narrow observation and action spaces imposed by their embodiments such as sensor configurations, end-effector design, actuation limits and coordinate conventions. As a result, robotic manipulation is largely formulated and benchmarked within these same narrow settings, with only weak support for richer embodied modalities. Moreover, because mechanical parameters, morphology, and control interfaces differ across platforms, the spaces they induce are not directly equivalent, making cross-embodiment alignment intrinsically difficult and potentially propagating these incompatibilities into downstream processes [29].

The open challenge, therefore, is to establish stronger embodied foundations for manipulation, which requires progress on three tightly coupled dimensions:

- **Cross-embodiment abstraction.** Shared interfaces and abstractions are needed to isolate downstream processes from hardware-specific incompatibilities.
- **Modality coverage.** Support for richer observation and action modalities needs to be broadened under the constraints imposed by heterogeneous embodiments.
- **Embodiment adaptation.** Shared abstractions also need principled mechanisms for adaptation to specific platforms, so that differences in morphology, sensing, actuation, and control interfaces can be handled systematically.

6.2 Data Ecosystem with Unified Support

Current data ecosystem for robotic arm manipulation is structurally insufficient. Real collection methods such as teleoperation provide the strongest coverage of all four components of $\mathcal{D}$, but are slow, expensive, and difficult to scale under real embodied interaction. More portable or autonomous alternatives improve efficiency, but at the cost of the contact events and state transitions that can only be captured through direct physical interaction. Simulation and synthetic data further improve scalability, but $\mathcal{D}$ they produce is only an incomplete proxy for real-world manipulation, especially where contact, friction, compliance, deformable dynamics, and cross-embodiment adaptation are central. As a result, $\mathcal{D}$ is often limited in scale and uneven in coverage across the components most critical to manipulation. Even when such data can be obtained, differences in control interfaces and platform conventions make it difficult to organize under shared schemas across platforms [26].

The open challenge, therefore, is to build a foundation for $\mathcal{D}$ that can sustain robotic arm manipulation reliably at scale, which requires progress on four tightly coupled dimensions:

- **Data collection.** The collection of $\mathcal{D}$ needs to become more scalable without sacrificing the fidelity of contact events and state transitions required by real-world manipulation.
- **Data standardization.** $\mathcal{D}$ needs to be organized under shared schemas and formats that enable consistent alignment across platforms and embodiments.
- **Data management.** $\mathcal{D}$ needs to be systematically curated, versioned, and made reusable across pipelines and research efforts.
- **Data alignment.** Beyond shared schemas and formats, $\mathcal{D}$ needs explicit alignment across embodiments, control interfaces, and task instances to support transfer across platforms.

6.3 Learning Paradigms with Stable Architectural Design

Current architectural design of learning paradigms for robotic arm manipulation is fundamentally unstable. The root cause lies in the lack of a principled and reusable definition of the action primitive. Without such a definition, cognitive planning (B) has no stable basis for determining what intermediate structures to produce, and policy learning (P) has no stable basis for determining what action granularity to operate on. As interfaces are specified independently across methods, the functional separation between B and P and the domain of applicability of different paradigms become progressively underdetermined [95]. This difficulty is further compounded by embodiment constraints and incomplete $\mathcal{K}$. Meanwhile, action representations diversify in the absence of a shared constraining basis, which in turn hinders the principled extension and cumulative refinement of existing paradigms.

The open challenge, therefore, is to build a stable architectural foundation for robotic arm manipulation around a consistent definition of embodied action, which requires progress on four tightly coupled dimensions:

- **Action representation.** Establish a shared executable action basis that defines minimal actionable units consistently across embodiments, viewpoints, and control regimes.
- **Temporal abstraction.** Develop principled mechanisms for composing, extending, and revising action primitives over time to coordinate planning and execution across tasks.
- **Functional decomposition.** Determine a principled division of labor between B and P based on a stable intermediate action representation.
- **System interfaces.** Establish reusable interface conventions that standardize how observations are encoded, how intermediate representations are passed across modules, and how different embodiments are adapted.

6.4 System Evaluation with Balanced Benchmarking

Current system evaluation for robotic arm manipulation is structurally imbalanced. Simulation supports controlled and reproducible comparison, but most benchmark suites still evaluate systems through task abstractions, observation conventions, and physics approximations that are only partial proxies for deployment. Real-world evaluation better reflects deployment conditions, but it is difficult to scale, standardize, and reproduce across embodiments, laboratories, and task setups [14]. As a small number of benchmark suites become widely adopted, their task definitions and evaluation protocols increasingly influence how progress in robotic arm manipulation is measured and compared. This imbalance is amplified by the architectural instability discussed above. When action interfaces, task decompositions, and success criteria are not stably defined, evaluation defaults to internally consistent but local proxies. Properties that are critical in deployment might be only weakly measured, including robustness under distribution shift, recovery from intermediate failure, execution efficiency, and long-horizon consistency. Binary task success further obscures where and why failure occurs, making evaluation a limited diagnostic signal for upstream design.

The open challenge, therefore, is to establish an evaluation foundation that is reproducible in controlled settings while still being informative about real deployment, which requires progress on four tightly coupled dimensions:

- **Shared evaluation abstractions.** Task definitions, action interfaces, and success criteria needs to be aligned so that comparisons can accurately reflect system capability.
- **Diagnostic measurement.** Evaluation needs to move beyond binary success to capture stagewise progress, recovery behavior, robustness to perturbations and distribution shift, execution efficiency, and safety margins, enabling failures to be localized and interpreted.
- **Scalable real-world protocols.** Real-world evaluation needs infrastructure for standardized resets, automatic logging and scoring, so that the same evaluation protocol can be repeated across embodiments and task setups with directly comparable results.
- **Failure evaluation.** Evaluation frameworks need to expose long-tail failures, recovery behavior, adversarial vulnerability, and other risks hidden by binary success metrics.

6.5 Trustworthiness Under Coupled Constraints

Trustworthiness emerges as an independent challenge once robotic arm manipulation systems are deployed in real-world settings. Strong task performance does not guarantee safety under rare conditions, security against malicious inputs, interpretability when failures occur, or reliable human oversight in shared environments. Because perception, reasoning, and action are tightly coupled, serious failure modes can hide until execution, while benchmark competence can coexist with substantial deployment risk. The difficulty is not only that failures occur, but that they are hard to expose, attribute, and control. Targeted triggers may induce control deviations without noticeably degrading benign-task performance [216]. At the same time, end-to-end policies provide limited visibility into how observations and instructions are translated into action, and even explicit intermediate representations may offer only a partial view of the decision process.

The open challenge, therefore, is to treat trustworthiness as a core requirement for robotic arm manipulation systems, which requires progress on four tightly coupled dimensions:

- **Operational safety.** Systems need mechanisms to prevent harmful actions, unsafe contact, collisions, force-limit violations, and other failures during real-world physical interaction.
- **Security robustness.** Manipulation policies need to resist malicious inputs, latent triggers, instruction perturbations, and backdoor vulnerabilities that can trigger unsafe control outputs.
- **Causal interpretability.** Methods are needed to decompose how observations, language, intermediate reasoning, memory, and action representations contribute to the final decision, so that failures can be attributed to their true source.
- **Calibrated human oversight.** Interaction paradigms need to make system intent, uncertainty, and control authority explicit to human operators, enabling anticipation, timely intervention, and clear operational boundaries in shared environments.

7 Conclusion

This article has examined robotic arm systems toward embodied intelligence by tracing the co-evolution of embodiment infrastructure, data ecosystems, learning paradigms, and evaluation practices. This perspective shows that recurring bottlenecks in robotic arm systems are rarely confined to a single layer. Challenges of robotic arm systems often arise from interactions among infrastructure, data, learning, and evaluation. Addressing these challenges therefore requires coordinated progress across the manipulation stack. We hope this paper provides a structured reference for future research and supports more systematic progress in robotic arm manipulation toward embodied intelligence.

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Received XXX; revised XXXX; accepted XXXX